

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

End-to-End Data Analytics Project Report

3.9K

Total Customers

\$233.08K

Total Revenue

\$59.76

Average Order Value

3.75

Average Review Rating

1. Project Overview

This project analyses customer shopping behaviour using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behaviour to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color).
 - Shopping behaviour (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Technology Stack

- Python (Pandas & Numpy)
- My SQL
- Power Bi
- Excel

4. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.

- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

```
[4]: df.describe(include='all')
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:**
 - Created **age_group** column by binning customer ages.
 - Created **purchase_frequency_days** column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis.

```
%pip install mysql-connector-python
```

```
%pip install sqlalchemy pymysql
```

```
%pip install pymysql sqlalchemy
```

```
from sqlalchemy import create_engine

# MySQL connection
username = "root"
password = "*****"
host = "localhost"
port = "3306"
database = "customer_behavior"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

# Write DataFrame to MySQL
table_name = "customer" # choose any table name
df.to_sql(table_name, engine, if_exists="replace", index=False)

# Read back sample
pd.read_sql("SELECT * FROM customer LIMIT 5;", engine)
```

5. Dataset & Methodology

Dataset overview

Attribute	Value
Records	3,900
Columns in cleaned workbook	19
Customer ID	1–3,900
Purchase amount	Numeric
Review rating	Decimal rating, 1–5
Customer demographics	Age, Gender, Location, Age Group
Purchase details	Item, Category, Size, Color, Season, Purchase Amount
Behavioral attributes	Subscription, Previous Purchases, Purchase Frequency
Operational attributes	Payment Method, Shipping Type, Discount Applied

Dataset Preparation

The project workflow began with loading and inspecting the dataset, checking structure and summary statistics, handling missing review-rating values, standardizing column names, and creating derived fields such as `age_group` and purchase-frequency information. The cleaned data was then used for SQL business analysis and Power BI visualization.

Analytical workflow

1	Data Loading	Load and inspect customer shopping data
2	Data Cleaning	Handle missing values and standardize fields
3	Feature Engineering	Create age groups and purchase-frequency features
4	SQL Analysis	Answer business questions with aggregations, CTEs and window functions
5	Power BI	Build interactive dashboard with KPIs, slicers and charts
6	Insights	Translate findings into business recommendations

6. Business Analysis

The following 15 SQL analyses form the final business-analysis layer of the project. The first ten are the core questions used in the original analysis, and Q11–Q15 extend the analysis into category performance, discount impact, payment methods, location and purchase frequency.

Customer & Revenue Analysis

Q1. What is the total revenue generated by male vs. female customers?

gender	Total_Revenue	In_Percentage
Male	157890	67.74
Female	75191	32.26

Finding: Male customers generated \$157,890 (67.73% of total revenue), while female customers generated \$75,191 (32.27%). The male group also has more transactions in the dataset

Q2. Which customers used a discount but still spent more than the average purchase amount?

Finding: There are 839 purchases where a discount was applied and the purchase amount exceeded the overall average of \$59.76. These purchases range from \$60 to \$100

Q3. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status	Total Customer	Average Purchase Amount	Total Revenue
▶	Yes	1053	59.492	62645
	No	2847	59.865	170436

Finding: Non-subscribers account for 2,847 transactions and \$170,436 revenue, while subscribers account for 1,053 transactions and \$62,645 revenue. In this dataset, subscribers do not have a higher average purchase value.

Q4. Which category performs best in terms of sales, revenue, and average purchase value?

	category	Total_sales	Total_Revenue	Avg_Purchase_amount
►	Clothing	1737	104264	60.03
	Accessories	1240	74200	59.84
	Footwear	599	36093	60.26
	Outerwear	324	18524	57.17

Finding: Clothing is the strongest category with 1,737 purchases and \$104,264 revenue. Accessories is second with \$74,200 revenue. Footwear has the highest average purchase value at \$60.26, while Outerwear has the lowest at \$57.17.

Q5. What is the revenue contribution of each age group?

	age_group	TOTAL_REVENUE
►	adult	50246
	young	48406
	teenager	48040
	middle age	43225
	senior	43164

Finding: Adults contribute the highest revenue (\$50,246), followed by Young (\$48,406) and Teenager (\$48,040). Middle Age and Senior groups contribute \$43,225 and \$43,164 respectively.

Product Analysis

Q1 Which are the top 5 products with the highest average review rating?

	item_purchased	Average Review rating
►	Gloves	3.861
	Sandals	3.844
	Boots	3.819
	Hat	3.801
	Skirt	3.785

Finding: Gloves has the highest average review rating among the products in the dataset, followed by Sandals, Boots, Hat and Skirt.

Q2. Which 5 products have the highest percentage of purchases with discounts applied?

	item_purchased	Percentage
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

Finding: Hat has the highest share of discounted purchases at 50.00%, followed by Sneakers (49.66%), Coat (49.07%), Sweater (48.17%) and Pants (47.37%).

Q3. What are the top 3 most purchased products within each category?

	RNK	category	item_purchased	TOTAL
▶	1	Accessories	Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

Finding: The category-level ranking highlights Jewellery, Belt and Sunglasses in Accessories; Blouse, Pants and Shirt in Clothing; Sandals, Shoes and Sneakers in Footwear; and Jacket and Coat among the leading Outerwear items. Outerwear has only two products in the top-three result because the dataset contains fewer distinct Outerwear items.

Q4. Do discounts actually increase the average purchase amount?

	discount_applied	Total_Purchases	Avg_Purchase_Amount	Total_Revenue
▶	Yes	1677	59.28	99411
	No	2223	60.13	133670

Finding: Non-discounted purchases have an average purchase value of \$60.13 versus \$59.28 for discounted purchases. The discounted group therefore has an average value about \$0.85 lower in this dataset

Customer Behaviour

Q1. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

	Customer_Segment	No._of_customer
▶	LOYAL	3116
	RETURNING	701
	NEW	83

Finding: The dataset is dominated by the Loyal segment: 3,116 customers (79.90%) are classified as Loyal, 701 (17.97%) as Returning and 83 (2.13%) as New under the project's stated thresholds

Q2. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	subscription_status	Repeat_Customers	In_Percentage
►	Yes	958	27.56
	No	2518	72.44

Finding: Among customers with more than five previous purchases, 27.56% are subscribed and 72.44% are not subscribed. In this dataset, repeat buying is therefore not strongly associated with subscription enrolment.

Q3. Does purchase frequency affect customer spending?

	frequency_of_purchases	Avg_Purchase_Amount	Total_Revenue
►	Bi-Weekly	60.69	33200
	Annually	60.17	34419
	Every 3 Months	60.08	35088
	Quarterly	59.98	33771
	Monthly	59.33	32810
	Fortnightly	59.05	32007
	Weekly	58.97	31786

Finding: Bi-Weekly customers have the highest average purchase value at \$60.69, while Weekly customers have the lowest at \$58.97. The difference is modest, suggesting purchase frequency alone is not a major driver of transaction value in this dataset.

Operations & Preferences

-- Q1. Compare the average Purchase Amounts between Standard and Express Shipping.

	shipping_type	Average Purchase Amount
►	Express	60.475
	Standard	58.460

Finding: Express-shipping purchases have a higher average purchase amount (\$60.48) than Standard-shipping purchases (\$58.46), a difference of about \$2.02

-- Q2. Which payment methods generate the most revenue?

	payment_method	Total_Transaction	Total_Revenue	In_Percentage
►	Credit Card	671	40310	17.29
	PayPal	677	40109	17.21
	Cash	670	40002	17.16
	Debit Card	636	38742	16.62
	Venmo	634	37374	16.03
	Bank Transfer	612	36544	15.68

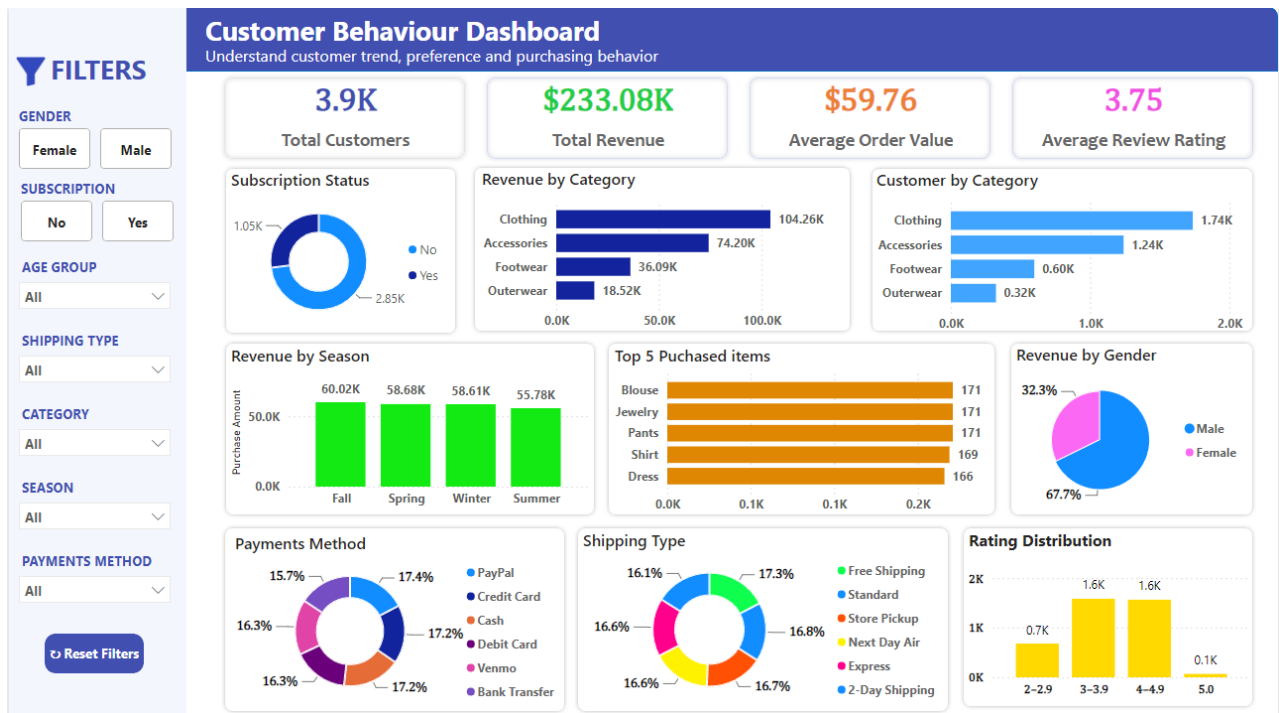
Finding: Credit Card generates the highest total revenue at \$40,310, narrowly ahead of PayPal at \$40,109. Debit Card has the highest average purchase value at \$60.92.

-- Q3. Which locations generate the most revenue?

	location	Total_Revenue	In_Percentage
►	Montana	5784	2.48
	Illinois	5617	2.41
	California	5605	2.40
	Idaho	5587	2.40
	Nevada	5514	2.37
	Alabama	5261	2.26
	New York	5257	2.26
	North Dakota	5220	2.24
	West Virginia	5174	2.22
	Nebraska	5172	2.22

Finding: Montana leads the top-10 location list with \$5,784 revenue, followed by Illinois (\$5,617) and California (\$5,605). Nevada has a relatively high average purchase value of \$63.38 despite having fewer transactions than several leading locations

6. Power BI Dashboard



Dashboard visuals

- **Subscription Status** — subscribed vs. non-subscribed customer counts.
- **Revenue by Category** — revenue contribution from Clothing, Accessories, Footwear and Outerwear.
- **Customer by Category** — number of customers/purchases by category.
- **Revenue by Season** — seasonal revenue comparison.
- **Top 5 Purchased Items** — most frequently purchased individual products.
- **Revenue by Gender** — revenue contribution by male and female customers.
- **Payment Method** — share of revenue/transactions across payment methods.
- **Shipping Type** — distribution across shipping options.
- **Rating Distribution** — review-rating bands for decimal ratings.

7. Dashboard Findings

- **Overall performance:** The dashboard reports 3.9K customers/transactions, \$233.08K total revenue, \$59.76 average order value and a 3.75 average review rating.
- **Subscription:** 1,053 customers are subscribed and 2,847 are not, so approximately 27% of the dataset is subscribed.
- **Category:** Clothing is the leading category at \$104.26K revenue and approximately 1.74K purchases.
- **Season:** Fall has the highest revenue at \$60.02K, followed by Spring (\$58.68K), Winter (\$58.61K) and Summer (\$55.78K).
- **Top products:** Blouse, Jewellery and Pants are the most purchased individual items at 171 purchases each, followed by Shirt (169) and Dress (166).
- **Gender:** Male customers contribute \$157.89K revenue compared with \$75.19K from female customers.
- **Payment:** Payment shares are relatively balanced; PayPal is the largest share in the dashboard, while SQL shows Credit Card produces the highest total revenue.
- **Shipping:** Shipping choices are distributed fairly evenly across the six shipping methods, with Free Shipping having the largest transaction count.
- **Ratings:** The rating distribution is concentrated in the 3–3.9 and 4–4.9 bands, with 1,586 and 1,566 reviews respectively; 68 reviews are exactly 5.0.

7. Business Recommendations

- **Prioritize Clothing:** Clothing generates the largest revenue and purchase volume. Maintain strong inventory availability and use top-selling items such as Blouse, Pants and Shirt in campaigns.
- **Improve Subscription Conversion:** Only about 27% of the dataset is subscribed, and repeat buyers are mostly non-subscribed. Loyalty rewards, exclusive benefits and targeted subscription offers could be used to convert repeat customers.
- **Review Discount Strategy:** Discounted transactions have a slightly lower average purchase value than non-discounted transactions. Evaluate whether discounts are driving incremental volume or simply reducing transaction value.
- **Promote High-Rated Products:** Products such as Gloves, Sandals and Boots have strong average ratings. These products can be highlighted in campaigns, recommendations and product-placement strategies.
- **Optimize Shipping Offers:** Express and 2-Day Shipping are associated with higher average purchase values than

- Standard Shipping. Premium delivery options can be positioned as upsell opportunities while monitoring cost and margin.
- Use Location-Level Targeting: Revenue varies across locations. Top-revenue locations can receive stronger marketing and inventory attention, while high-average-value locations can be targeted with premium offers.
- Target by Customer Segment: The dataset is heavily weighted toward Loyal customers under the project's segmentation rule. Retention campaigns should protect this base while targeted onboarding campaigns can increase the New and Returning segments.

8. Project Limitations & Notes

- The dataset represents transaction-level customer shopping behaviour and does not provide profit, cost, margin or customer lifetime value, so revenue findings should not be interpreted as profitability findings.
- Purchase frequency and previous purchases are fields in the dataset rather than a full longitudinal customer history; therefore, behavioural relationships should be interpreted as patterns within the provided data.
- The dashboard uses aggregated visuals. A dashboard metric can change when slicers are applied, while the SQL findings above represent the full dataset unless the query explicitly applies a filter.
- The customer segmentation thresholds are project-defined: 1 previous purchase = New, 2-10 = Returning, and more than 10 = Loyal.

9. Conclusion

This project demonstrates an end-to-end data analytics workflow: preparing customer shopping data, transforming it into analysis-ready fields, answering business questions with SQL, and presenting the results through an interactive Power BI dashboard. The final analysis identifies the strongest categories and products, spending differences across customer groups, discount behaviour, subscription patterns, shipping and payment preferences, customer ratings, and location-level revenue. The combination of technical analysis and business recommendations makes the project suitable as a Data Analyst portfolio case study.